ECE 302
Prof. Ilya Pollak
Final Exam Solutions, 5/4/2012, 3:20-5:20pm in LILY 1105.

- Your ID will be checked during the exam.
- Please bring a No. 2 pencil to fill out the answer sheet.
- This is a closed-book exam. No calculators are allowed. No electronic devices of any kind are allowed. A formula sheet is provided on the back of this page.
- **Please read the instructions on this page very carefully.**
- Please fill out the following information on your answer sheet:
  - `Instructor:` Ilya Pollak
  - `Course:` ECE 302
  - `Date:` 05/04/2012
  - sign the answer sheet
  - fill in your last name and the first six letters of your first name
  - fill in your student ID
  - fill in your section number: **0001 if you are in the 3:30-4:20 section and 0002 if you are in the 12:30-1:20 section**
- You have 2 hours to complete 20 multiple-choice questions.
- Each correct answer is worth one point, for a total of 20 points.
- After the first student turns in his/her answer sheet and leaves, no one will be allowed to temporarily leave the exam room and come back, unless it is an emergency.
- Please only turn in the answer sheets.
- There will be no partial credit.
- **This document is double-sided.**

Random variable	PMF or PDF	Mean	Variance
Bernoulli	p for $k=1$; $1-p$ for $k=0$.	p	$p(1-p)$
Discrete uniform	$\frac{1}{n}$, $k=k_0+1, k_0+2, \ldots, k_0+n$	$k_0+\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Geometric	$(1-p)^{k-1}p$, $k=1,2,3,\ldots$	$\frac{1}{p}$	$\frac{1}{p^2}-\frac{1}{p}$
Binomial	$\left(\begin{array}{c} n \\ k \end{array}\right)(1-p)^{n-k}p^k$, $k=0,1,\ldots,n$	pn	$np(1-p)$
Continuous uniform	$\frac{1}{b-a}$, $a \leq x \leq b$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Shifted exponential	$\lambda e^{-\lambda(x-\alpha)}$, $x \geq \alpha$	$\lambda^{-1}+\alpha$	λ^{-2}
Two-sided exponential	$\begin{cases} p\lambda e^{-\lambda x}, & x \geq 0 \\ (1-p)\lambda e^{\lambda x}, & x < 0 \end{cases}$	$(2p-1)\lambda^{-1}$	$(1+4p-4p^2)\lambda^{-2}$
Erlang	$\dfrac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}$, $x \geq 0$	$k\lambda^{-1}$	$k\lambda^{-2}$
Normal (Gaussian)	$\dfrac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Lognormal	$\dfrac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$, $x>0$	$e^{\mu+\sigma^2/2}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$
Cauchy	$\dfrac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	undefined	undefined
Rayleigh	$\dfrac{1}{\sigma^2} x e^{-\frac{x^2}{2\sigma^2}}$, $x \geq 0$	$\sigma\sqrt{\frac{\pi}{2}}$	$\frac{4-\pi}{2}\sigma^2$
Pareto	$a\frac{c^a}{x^{a+1}}$, $x \geq c$,	$\frac{ac}{a-1}$, $a>1$	$\frac{ac^2}{(a-1)^2(a-2)}$, $a>2$

where

$$\left(\begin{array}{c} n \\ k \end{array}\right) = \frac{n!}{(n-k)!k!}.$$

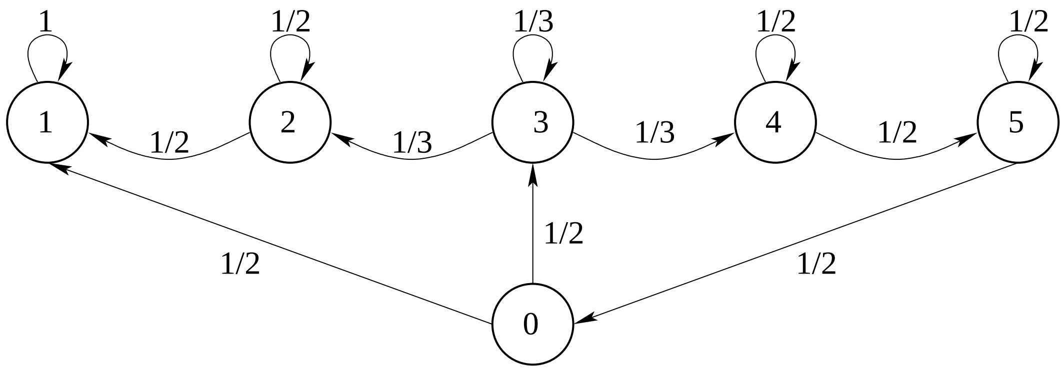

Figure 1: Markov chain for Questions 1–3.

Questions 1–3. Consider the Markov process pictured in Fig. 1. Given that this process is in state 0 just before the first transition, determine the following probabilities.

Question 1. The probability that the process enters state 4 for the first time as a result of the 6-th transition.
Question 2. The probability that the process never enters state 4.
Question 3. The probability that the process never enters state 2.

For each of the Questions 1–3, choose the answer from among the following options (the three answers are not necessarily all different!):

1. 1
2. 5/6
3. 3/4
4. 2/3
5. 1/2
6. 1/3
7. $1/(3^5)$
8. $1/(2 \cdot 3^5)$
9. impossible to determine based on the given information
10. none of the above

Solution. Starting from 0, the only way to reach 4 for the first time after six steps is

$$0 \to 3 \to 3 \to 3 \to 3 \to 3 \to 4$$

The probability of this is:

$$P_K = p_{03} p_{33}^4 p_{34} = (1/2)(1/3)^4(1/3) = \frac{1}{2 \cdot 3^5} = 1/486.$$

Therefore, the answer key for Question 1 is 8.

If the process goes directly from 0 to 1, it will never enter state 4. The probability of this is $p_{01} = 1/2$. If the first transition is from zero to three, then it will never get to 4 under one of the following two scenarios: (i) if it stays at 3 forever—the probability of this is zero; (ii) if it will eventually go to 2. Since $p_{32} = p_{34}$, the conditional probability that the process will eventually go to 2 is equal to its conditional probability to go to 4—i.e., both probabilities are 1/2. Therefore, the probability to go from 0 to 3 and the eventually to 2 is $(1/2)(1/2) = 1/4$. The total probability that the process never enters state 4 is then $1/4 + p_{01} = 3/4$. Therefore, the answer key for Question 2 is 3.

Note that the process can only go from 0 to 1 or to 3. If it goes to 1, it will never get to 2. If it goes to 3, it will either eventually get to 2 or to 4, with equal conditional probabilities of 1/2 each. If from 3 the process eventually gets to 4, it will eventually get back to 0, with probability 1. Therefore, {never getting from 0 to 2} is equivalent to $\{\{0 \to 1\}$ or $\{0 \to 3$, followed by eventually getting back to 0, followed by never getting from 0 to 2$\}\}$. Denoting by P the conditional probability to never get from 0 to 2 given that the process is at 0, we therefore have: $P = 1/2 + (1/2) \cdot (1/2) \cdot P$, or $P = 2/3$. The answer key for Question 3 is 4.
Question 1 answer key: 8.
Question 2 answer key: 3.
Question 3 answer key: 4.

Questions 4–5. Consider a three-state Markov chain, with states 0, 1, and 2. The following one-step transition probabilities are given: $p_{10} = 1 - p$, $p_{11} = 0$, $p_{21} = 1$, and $p_{01} = 1$, where p is a real number such that $0 < p < 1$.

Question 4. Given that the process is in state 2 immediately before the first transition, what is the probability that it is in state 0 after 2012 transitions?

Question 5. Let $r_{20}(n)$ be the n-step transition probability from state 2 to state 0. What is $\lim_{n\to\infty} r_{20}(n)$?

For each of these two questions, choose the answer from among the following options (the two answers are not necessarily different!):

1. 1
2. p
3. $1 - p$
4. $p(1-p)^{2011}$
5. $p^{2011}(1-p)$
6. $1 + p + p^2 + ... + p^{2011}$
7. 1/2
8. 0
9. 1/4
10. none of the above

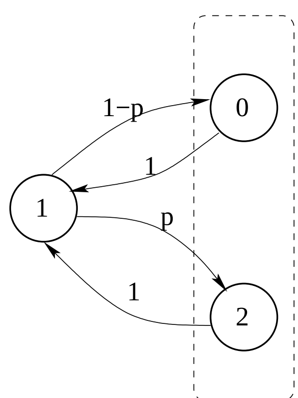

Figure 2: Markov chain for Questions 4–5.

Solution. The Markov chain described in the statement of the problem is depicted in Fig. 2. It has a single recurrent class, and is periodic with period 2. From state 1, the process must go to the set of states $\{0, 2\}$ (inside the dashed line in the figure), and from state 0 or 2 it must go to 1. Therefore, having started at 2, the process is in state 1 after any odd number of steps, such as 2011. One step later, the process is in state 0 with probability $1 - p$ and in state 2 with probability p. Therefore, the answer to Question 4 is $1 - p$ (answer key 3).

Because of the periodicity, the n-step transition probabilities do not converge. For odd n, $r_{20}(n) = 0$. For even n, $r_{20} = 1 - p \neq 0$, since we are given than $p \neq 1$. Therefore, the limit does not exist, and the answer key for Question 5 is 10.

Question 4 answer key: 3.
Question 5 answer key: 10.

Questions 6–7. Consider a two-state Markov chain, with states 1 and 2. The following one-step transition probabilities are given: $p_{12} = p_{22} = 1/3$.

Question 6. Given that the process is in state 1 immediately before the first transition, what is the probability that it is in state 1 after 5 transitions?

Question 7. Let $r_{22}(n)$ be the n-step transition probability from state 2 to state 2. What is $\lim_{n\to\infty} r_{22}(n)$?

For each of these two questions, choose the answer from among the following options (the two answers are not necessarily different!):

1. 0
2. 1/6
3. 1/4
4. 1/3
5. 1/2
6. 2/3
7. $(1/3)^5$
8. $(2/3)^5$
9. the answer cannot be determined based on the given information
10. none of the above

Solution. Note that the Chapman-Kolmogorov equations,

$$\begin{aligned}
r_{11}(n) &= r_{11}(n-1)p_{11} + r_{12}(n-1)p_{21}, \\
r_{12}(n) &= r_{11}(n-1)p_{12} + r_{12}(n-1)p_{22}, \\
r_{22}(n) &= r_{22}(n-1)p_{22} + r_{21}(n-1)p_{12}, \\
r_{21}(n) &= r_{22}(n-1)p_{21} + r_{21}(n-1)p_{11},
\end{aligned}$$

are all satisfied by $r_{11}(n) = r_{21}(n) = 2/3$ and $r_{12}(n) = r_{22}(n) = 1/3$ for all positive integer n. Therefore, the process in steady state after the first step, and $r_{11}(5) = 2/3$, and the steady-state probability for state 2 is 1/3.

Question 6 answer key: 6.
Question 7 answer key: 4.

Question 8. In a communication channel, each transmitted symbol is a random variable which can assume four values, a_1, a_2, a_3, and a_4, with the following probabilities: $\mathbf{P}(a_1) = \mathbf{P}(a_4) = \frac{3}{8}$ and $\mathbf{P}(a_2) = \mathbf{P}(a_3) = \frac{1}{8}$. Consider the following five codes for this random variable.

a_1	a_2	a_3	a_4
001	010	011	100

- Code (A)

a_1	a_2	a_3	a_4
01	10	11	10

- Code (B)

a_1	a_2	a_3	a_4
00	010	011	1

- Code (C)

a_1	a_2	a_3	a_4
0	100	101	11

- Code (D)

a_1	a_2	a_3	a_4
11	100	10	0

- Code (E)

Which of these codes are optimal (in the sense of minimizing the expected codeword length) uniquely decodable codes?

1. Only (A).
2. Only (B)
3. Only (E).
4. Only (A) and (B).
5. Only (B) and (C).
6. Only (C) and (D).
7. Only (D) and (E).
8. Only (C), (D), and (E).
9. None of these five codes are optimal uniquely decodable codes.
10. None of the above.

Solution. Codes (C) and (D) are both Huffman codes and hence are optimal and uniquely decodable. Their expected codeword length is $(3/8) \cdot (1 + 2) + 2 \cdot (1/8) \cdot 3 = 15/8$. Codes (A) and (B) are fixed-length codes with expected codeword lengths 3 and 2, respectively. They are not optimal since both 3 and 2 are larger than 15/8. Code (E) has a smaller expected codeword length than (C) and (D) but is not uniquely decodable, since "100" could mean either a_2 or a_3a_4.
Answer key: 6.

Questions 9–10.
Question 9. Find the entropy of a geometric random variable with parameter 0.5.
Question 10. X is a discrete uniform random variable with outcomes $1, 2, \ldots, 1024$. Find the entropy of X.

For each of these questions, choose the answer from among the following options (the two answers are not necessarily different!):

1. 1
2. 2
3. 3
4. 4
5. 5
6. 6
7. 7
8. 8
9. 9
10. 10

Solution. A geometric random variable has positive integer outcomes. The probability of any outcome k is 2^{-k}, and its self-information is therefore $-\log_2 2^{-k} = k$. Therefore, the expected self-information is equal to the expectation of the geometric random variable itself, 2. The entropy is, by definition, the expected self-information. The answer key for Question 9 is therefore 2.

In Question 10, the probability of each outcome is 2^{-10} and therefore the self-information of each outcome is $-\log_2 2^{-10} = 10$. The entropy is the expected self-information, which is also 10. The answer key for Question 10 is therefore 10.
Question 9 answer key: 2.
Question 10 answer key: 10.

Question 11. X is a binomial random variable with parameters 0.5 and 4, and C is an optimal uniquely decodable code for X. (*Optimal* here means that C minimizes the expected codeword length among all the uniquely decodable codes for X.) Find the expected codeword length for C.

1. 1
2. 2
3. 3
4. 4
5. $21/8 - (3/8)\log_2 3$
6. $29/8 - (3/8)\log_2 3$
7. 5/2
8. 9/4
9. 17/8
10. None of the above

Solution. The possible values for X are 0, 1, 2, 3, and 4, with respective probabilities 1/16, 4/16, 6/16, 4/16, and 1/16. Recursively combining two least likely symbols yields the following Huffman tree (internal nodes are marked with x's, and corresponding probabilities are in brackets):

```
          [2/16]      [6/16]     [10/16]
0 -------- x -------- x-------- x -------- x
          /          /          /          /
         /          /          /          /
        4          1          3          2
```

The resulting Huffman code is: 0000 for 0, 001 for 1, 1 for 2, 01 for 3, and 0001 for 4. Its expected codeword length is $4(1/16) + 3(4/16) + 6/16 + 2(4/16) + 4(1/16) = 34/16 = 17/8$. Note that there are other Huffman trees resulting in other Huffman codes, for example, $\{000, 01, 10, 11, 001\}$, but they all have the same expected codeword length.
Answer key: 9.

Question 12. Let $X_1, X_2, \ldots$ be a sequence of discrete independent, identically distributed random variables with mean $E[X_i] = 0$ and variance $\text{var}(X_i) = 1$ for all positive integer i. Further, let $Y_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ for all positive integer n. Consider the following inequality:

$$\mathbf{P}(|Y_n| \geq 1) > 0.01.$$

Only one of the following statements is correct. Select the correct statement.

1. The inequality is satisfied for $n = 100$.
2. The inequality is satisfied for all integer n such that $n \geq 200$.
3. The inequality is satisfied for all integer n such that $n \geq 100$.
4. The inequality is satisfied for all positive integer n such that $n < 200$.
5. The inequality is satisfied for all integer n such that $1 \leq n \leq 100$.
6. The inequality is satisfied for all integer n such that $10 \leq n \leq 100$.
7. The inequality is satisfied for all integer n such that $90 \leq n \leq 100$.
8. The inequality is satisfied for all positive integer n such that $n \leq 1000$.
9. The inequality cannot be satisfied for any positive integer n.
10. None of the above.

Solution. Note that $E[Y_n] = 0$, and $\text{var}(Y_n) = 1/n$. Using Chebyshev's inequality for random variable Y_n and choosing $c = 1$ we get $\mathbf{P}(|Y_n - 0| \geq 1) \leq \frac{1}{n}$. Therefore, the inequality

$$\mathbf{P}(|Y_n| \geq 1) > 0.01$$

is violated for any $n \geq 100$. Options 1-8 all include cases $n \geq 100$ and therefore are false. Option 9 is also false. To show this, we need one example of $X_1, X_2, \ldots, X_n$ and n such that the given inequality is satisfied. Let X_1 be a discrete random variable such that $p_{X_1}(1) = p_{X_1}(-1) = 1/2$. Then $E[X_1] = 0$ and $\text{var}(X_1) = E[X_1^2] = 1^2 \cdot (1/2) + (-1)^2 \cdot (1/2) = 1$. For $n = 1$, we have $Y_n = X_1$, and $\mathbf{P}(|Y_n| \geq 1) = \mathbf{P}(|X_1| \geq 1) = 1 > 0.01$. So options 1-9 are all false, and therefore option 10 is true.
Answer key: 10.

Question 13. $X_1, X_2, \ldots$ are independent Bernoulli random variables with parameter $p = 1/3$.

$$
\begin{aligned}
Y_n &= \frac{1}{\sqrt{n}}(X_1 + X_2 + \ldots + X_n) \text{ for all positive integer } n, \\
W_n &= \frac{1}{\sqrt{n}}(X_1 + X_2 + \ldots + X_n - n/3) \text{ for all positive integer } n, \\
V_n &= X_1 + X_2 + \ldots + X_n - n/3 \text{ for all positive integer } n.
\end{aligned}
$$

The CDFs of Y_n, W_n, and V_n are F_{Y_n}, F_{W_n}, and F_{V_n}, respectively. Consider the following three statements.

A. The central limit theorem implies that there exists a normal CDF F such that $\lim_{n\to\infty} F_{Y_n}(y) = F(y)$ for all real y.

B. The central limit theorem implies that there exists a normal CDF G such that $\lim_{n\to\infty} F_{W_n}(w) = G(w)$ for all real w.

C. The central limit theorem implies that there exists a normal CDF H such that $\lim_{n\to\infty} F_{V_n}(v) = H(v)$ for all real v.

Select one of the following.

1. A, B, and C are all false.
2. A is true; B and C are false.
3. B is true; A and C are false.
4. C is true; A and B are false.
5. A and B are true; C is false.
6. A and C are true; B is false.
7. B and C are true; A is false.
8. A, B, and C are all true.
9. It is impossible to determine whether A, B, and C are true or false.
10. None of the above.

Solution. Since $E[X_n] = 1/3$, and $\text{var}(X_n) = 2/9$ we have that $E[Y_n] = \sqrt{n}/3$ and $\text{var}(Y_n) = 2/9$. Therefore, $F_{Y_n}(y) \to 0$ for any y, and so A is wrong. The variance of V_n is $2n/9$, and therefore $F_{V_n}(v) \to 0$ for any v, and so C is wrong. However, $E[W_n] = 0$ and $\text{var}(W_n) = 2/9$, and therefore the central limit theorem is applicable to W_n's. B is true.
Answer key: 3.

Question 14. For any integer $n \geq 2$, X_n is a Bernoulli random variable with parameter $p = 1/n$. Random variables $X_2, X_3, \ldots$ are independent. For any integer $n \geq 2$, random variable Y_n is defined as $Y_n = nX_n$. Consider the following four statements.

A. There exists a real number a such that $\lim\limits_{n\to\infty} E[(Y_n - a)^2] = 0$.

B. There exists a real number b such that $\lim\limits_{n\to\infty} E[Y_n] = b$.

C. There exists a real number c such that the sequence $Y_2, Y_3, \ldots$ converges to c in probability.

D. There exists a real number d such that the sequence $Y_2, Y_3, \ldots$ converges to d with probability 1.

Select one of the following.

1. A, B, C, and D are all false.
2. A is true; B, C, and D are false.
3. B is true; A, C, and D are false.
4. C is true; A, B, and D are false.
5. D is true; A, B, and C are false.
6. A and B are true; C and D are false.
7. A and C are true; B and D are false.
8. B and C are true; A and D are false.
9. B and D are true; A and C are false.
10. A, B, C, and D are all true.

Solution. Since $E[X_n] = E[X_n^2] = 1/n$, we have that $E[Y_n] = 1$ and $E[Y_n^2] = n^2 E[X_n^2] = n$. Therefore, B is correct, with $b = 1$. For A, we have that

$$E[(Y_n - a)^2] = E[Y_n^2] - 2aE[Y_n] + a^2 = n - 2a + a^2.$$

This diverges for any a, and therefore A is false.

To show convergence in probability, note that, for any $\varepsilon > 0$, we have $\mathbf{P}(|Y_n| \geq \varepsilon) \leq 1/n \to 0$. Therefore, the sequence converges to zero in probability, and C is true with $c = 0$.

Since the n-th term of the sequence is either 0 or n, the only value that the sequence could potentially converge to with probability 1 is zero. In order for this convergence to take place, the following event would have to have probability 1: that, starting with some index k, all the terms in the sequence are zero. It was shown in class that this is not the case. Therefore, the sequence does not converge with probability 1, and D is false.

So, A and D are false, and B and C are true.
Answer key: 8.

Question 15. The probability density function of random variable X_i, for any positive integer i, is

$$f_{X_i}(x) = \begin{cases} \frac{1}{x} & \text{if } 1 \le x \le e \\ 0 & \text{otherwise} \end{cases}$$

Random variables $X_1, X_2, \dots$ are independent. Let $Y_n = (X_1 \cdot X_2 \cdot \ldots \cdot X_n)^{1/n}$, for any positive integer n. Does the sequence $Y_1, Y_2, \dots$ converge with probability 1? If so, select one of the answer keys 1–8; otherwise, select answer key 9. If there is not enough information provided to either decide whether the sequence converges with probability 1 or to determine the limit, select answer key 10.

1. $Y_n \to 1$ with probability 1.
2. $Y_n \to e^{0.5}$ with probability 1.
3. $Y_n \to e^2$ with probability 1.
4. $Y_n \to e^{-1}$ with probability 1.
5. $Y_n \to e$ with probability 1.
6. $Y_n \to 0$ with probability 1.
7. $Y_n \to -\ln(2)$ with probability 1.
8. The limit exists, but it is not listed in the answer keys 1–7.
9. The limit does not exist.
10. The given information is not enough to determine the solution.

Solution. In order to use any of the limit theorems we considered in the course, we would need to tie Y_n's to sums of independent random variables rather than products. We can convert a product into a sum by using logarithms:

$$\ln Y_n = \frac{1}{n} \sum_{i=1}^{n} \ln X_i.$$

Therefore, $\ln Y_n$ is the sample mean of $\ln X_1, \dots, \ln X_n$. Since $X_1, \dots, X_n$ are independent and identically distributed, so are $\ln X_1, \dots, \ln X_n$. In order to be able to use the strong law of large numbers, we would have to find the expectation of $\ln X_i$ and show that the variance of $\ln X_i$ is finite. First, let's calculate the expectation, using the change of variable $u = \ln x$:

$$E\left[\ln X_i\right] = \int_1^e \ln x \frac{1}{x} dx = \int_0^1 u du = \left. \frac{u^2}{2} \right|_0^1 = \frac{1}{2}.$$

Using the same change of variable, we have:

$$E\left[(\ln X_i)^2\right] = \int_1^e (\ln x)^2 \frac{1}{x} dx = \int_0^1 u^2 du = \left. \frac{u^3}{3} \right|_0^1 = \frac{1}{3}.$$

Therefore, $\ln X_i$ has finite variance, and we can apply the strong law of large numbers to the sequence $\ln X_1, \ln X_2, \ldots$. The sample means of this sequence, $\ln Y_n$, converge to the actual mean, $1/2$, with probability 1:

$$\mathbf{P}\left(\lim_{n\to\infty} \ln Y_n = \frac{1}{2}\right) = 1.$$

Therefore,

$$\mathbf{P}\left(\lim_{n\to\infty} Y_n = e^{0.5}\right) = 1.$$

Answer key: 2.

Question 16. This question deals with the same random variables as the previous question, but a different concept of convergence. The probability density function of random variable X_i, for any positive integer i, is

$$f_{X_i}(x) = \begin{cases} \frac{1}{x} & \text{if } 1 \leq x \leq e \\ 0 & \text{otherwise} \end{cases}$$

Random variables $X_1, X_2, \ldots$ are independent. Let $Y_n = (X_1 \cdot X_2 \cdot \ldots \cdot X_n)^{1/n}$, for any positive integer n. Does the sequence $Y_1, Y_2, \ldots$ converge in probability? If so, select one of the answer keys 1–8; otherwise, select answer key 9. If there is not enough information provided to either decide whether the sequence converges in probability or to determine the limit, select answer key 10.

1. $Y_n \to 1$ in probability.
2. $Y_n \to e^{0.5}$ in probability.
3. $Y_n \to e^2$ in probability.
4. $Y_n \to e^{-1}$ in probability.
5. $Y_n \to e$ in probability.
6. $Y_n \to 0$ in probability.
7. $Y_n \to -\ln(2)$ in probability.
8. The limit exists, but it is not listed in the answer keys 1–7.
9. The limit does not exist.
10. The given information is not enough to determine the solution.

Solution. It was shown in the previous question that this sequence converges to $e^{0.5}$ with probability 1. This implies that it also converges to $e^{0.5}$ in probability.
Answer key: 2.

Question 17. Continuous random variables X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} 1/4 & \text{if } |x| + |y| \leq \sqrt{2}, \\ 0 & \text{otherwise.} \end{cases}$$

A two-bit vector quantizer is used to jointly quantize X and Y, to obtain discrete random variables $\hat{X}$ and $\hat{Y}$. The quantizer is specified by four quantization sets in the plane, A_1, A_2, A_3, and A_4, and four quantization points (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4), such that the output of the quantizer is

$$(\hat{X}, \hat{Y}) = (x_i, y_i) \text{ if } (X, Y) \in A_i, \text{ for } i = 1, 2, 3, 4.$$

Among the following ten numbers, choose the smallest realizable mean-square error for such a two-bit quantizer. (For example, if two-bit quantizers can be constructed to achieve mean-square errors in keys 6-10 and cannot be constructed to achieve mean-square errors in keys 1-5, then your answer should be key 6. In other words, you must find the lowest possible mean-square error among the ten answer keys given to you. This is not necessarily the same as finding the lowest mean-square error among all possible quantizers.) The mean-square error is defined as $E[(X - \hat{X})^2 + (Y - \hat{Y})^2]$.

1. 1/6
2. 2/6
3. 3/6
4. 4/6
5. 5/6
6. 6/6
7. 7/6
8. 8/6
9. 9/6
10. 10/6

Solution. Let S' be a square centered at the origin with side length 2, and whose sides are parallel to the coordinate axes. Let random variables X' and Y' be jointly uniform over this square. Suppose that both X' and Y' are quantized using scalar one-bit uniform quantizers. Then the quantization bins are $[-1, 0]$ and $[0, 1]$ and the corresponding quantization levels are -0.5 and 0.5, for each of these two random variables. The overall mean-square error for this joint quantization strategy is

$$\begin{aligned} E[(X' - \hat{X}')^2 + (Y' - \hat{Y}')^2] &= 2E[(X' - \hat{X}')^2] \\ &= 2\left(\int_{-1}^{0} (x' + 0.5)^2 f_{X'}(x')\, dx' + \int_{0}^{1} (x' - 0.5)^2 f_{X'}(x')\, dx'\right) \\ &= 2\left(\int_{0}^{1} (x' - 0.5)^2\, dx'\right), \end{aligned}$$

where the last equality is obtained from symmetry and using the fact that the density of X' is 0.5 over the interval $[-1, 1]$. What's inside the parentheses is the variance of a continuous random variable which is uniformly distributed between 0 and 1. This is 1/12. Therefore, the mean-square error is 1/6.

The square S over which X and Y are uniform is obtained from rotating S' by 45 degrees. Since rotation does not change mean-square errors, the mean-square error of 1/6 is achievable. (Specifically, the quantizer that achieves it is a 45-degree rotation of the uniform quantizer for X' and Y'.) Since all the other answer keys are numbers above 1/6, 1/6 is the lowest achievable mean-square error among the given possibilities.
Answer key: 1.

Question 18. Find the mean-square error of the one-bit Lloyd-Max quantizer for a two-sided exponential random variable with parameters $p = 1/2$ and $\lambda = 1/3$. (One-bit quantizer has two quantization levels.) You can use the fact that the solution to the Lloyd-Max quantizer equations is unique for this problem.

1. 18
2. $\sqrt{18}$
3. 3
4. 4.5
5. $\sqrt{4.5}$
6. 6
7. $\sqrt{6}$
8. 8
9. 9
10. None of the above

Solution. Denote the random variable by X. Because of symmetry, we guess $x = 0$ to be the threshold for the Lloyd-Max quantizer. Then, in order to satisfy the Lloyd-Max quantizer equations derived in class, the two quantization levels must be the conditional means $E[X|X \geq 0]$ and $E[X|X \leq 0]$. The conditional density of X given $X \geq 0$ is exponential with parameter λ, and hence the conditional mean is $E[X|X \geq 0] = \lambda^{-1}$. The conditional density of $-X$ given $X \leq 0$ is exponential with parameter λ, and hence the conditional mean is $E[X|X \geq 0] = -\lambda^{-1}$. Note that all the Lloyd-Max quantizer equations are satisfied, because the quantization threshold $x = 0$ is the average of the two quantization levels.

Because of symmetry, the MSE over the entire real line is equal to twice the MSE over one of the two quantization intervals:

$$\text{MSE} = 2\int_0^\infty (x - \lambda^{-1})^2 f_X(x)\ dx.$$

This is the variance of an exponential random variable with parameter λ, and is therefore equal to λ^{-2}. For $\lambda = 1/3$, this is 9.
Answer key: 9.

If we weren't told that the solution is unique, we would need to verify this—i.e., that $x = 0$ is the only possible solution for the Lloyd-Max quantizer. Suppose that this is not the case, and suppose that $x > 0$. (The case $x \leq 0$ is handled similarly.) Then, based on what we know about shifted exponential random variables, $E[X|X \geq x] = x + \lambda^{-1}$. Recall that, according to the quantizer equations derived in class,

$$\frac{1}{2}(E[X|X \geq x] + E[X|X \leq x]) = x,$$

and therefore we must have $E[X|X \leq x] = x - \lambda^{-1}$. Denote

$$\mathbf{P}(X \geq x) = 0.5e^{-\lambda x} = q(x).$$

Note that $E[X] = 0$, but, on the other hand, using the total expectation theorem,

$$\begin{aligned} 0 &= E[X] = \mathbf{P}(X \leq x)E[X|X \leq x] + \mathbf{P}(X \geq x)E[X|X \geq x] \\ &= (1 - q(x))(x - \lambda^{-1}) + q(x)(x + \lambda^{-1}) = x - \frac{1 - 2q(x)}{\lambda} \\ &= x - \frac{1 - e^{-\lambda x}}{\lambda}. \end{aligned}$$

But this equation does not have any positive roots. To see this, note that the lines $y_1(x) = x$ and $y_2(x) = \lambda^{-1}(1 - e^{-\lambda x})$ touch at zero (since $y_1(0) = y_2(0) = 0$ and $y_1'(0) = y_2'(0) = 1$), and that $y_2'(x) < 1$ for $x > 0$ which means that these two lines never intersect for $x > 0$.

Therefore, the unique Lloyd-Max quantizer is the one described above—i.e., with threshold 0, quantization levels ± 3, and MSE 9.

Questions 19-20. Continuous random variable X has the following PDF:

$$f_X(x) = \begin{cases} x/2, & \text{if } 0 \le x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Discrete random variable Y is obtained by quantizing X as follows::

$$Y = \begin{cases} q_1, & \text{if } X \le x_1, \\ q_2, & \text{if } X > x_1. \end{cases}$$

Find the values q_1, x_1, and q_2 which minimize the expected absolute error $E[|X - Y|]$.

Question 19. Select one of the following.

1. $x_1 = 1$
2. $x_1 = 4/3$
3. $x_1 = \sqrt{2}$
4. $x_1 = 3/2$
5. $x_1 = \frac{\sqrt{2}}{4-\sqrt{3}}$
6. $x_1 = \frac{\sqrt{2}}{\sqrt{4-\sqrt{3}}}$
7. $x_1 = \frac{\sqrt{2}}{4-2\sqrt{2}}$
8. $x_1 = \frac{\sqrt{2}}{\sqrt{4-2\sqrt{2}}}$
9. Based on the given information, it is impossible to uniquely determine x_1.
10. None of the above.

Question 20. Select one of the following.

1. $q_1 = 1/2$ and $q_2 = 3/2$
2. $q_1 = 2/3$ and $q_2 = 5/3$
3. $q_1 = 1$ and $q_2 = 5/3$
4. $q_1 = 5/4$ and $q_2 = 7/4$
5. $q_1 = \frac{1}{\sqrt{4-2\sqrt{2}}}$ and $q_2 = \frac{2\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}}$
6. $q_1 = \frac{1}{\sqrt{4-\sqrt{3}}}$ and $q_2 = \frac{2\sqrt{2}-1}{\sqrt{4-\sqrt{3}}}$
7. $q_1 = 1$ and $q_2 = 2\sqrt{2} - 1$
8. $q_1 = \frac{1}{4-2\sqrt{2}}$ and $q_2 = \frac{2\sqrt{2}-1}{4-2\sqrt{2}}$
9. Based on the given information, it is impossible to uniquely determine q_1 and q_2.
10. None of the above.

Solution. Instead of just solving the given problem, we derive the general equations for an L-level scalar quantizer that minimizes the expected absolute error. We assume that the random variable X to be quantized has PDF f_X which is nonzero on some interval $a < x < b$ and zero outside of this interval. Here, a could be $-\infty$ and b could be $+\infty$. In the above question, $a = 0$ and $b = 2$. We use the same steps used in class to derive the equations for the scalar Lloyd-Max quantizer: we express the expected absolute error as a function of the endpoints of the quantization bins and the quantization levels, and differentiate to find the minimum.

$$
\begin{aligned}
E[|X-Y|] &= \sum_{k=1}^{L}\left[\int_{x_k}^{q_k}(q_k - x)f_X(x)\,dx + \int_{q_k}^{x_{k+1}}(x-q_k)f_X(x)\,dx\right] \\
&= \sum_{k=1}^{L}\left[q_k\int_{x_k}^{q_k} f_X(x)\,dx - \int_{x_k}^{q_k} xf_X(x)\,dx - \int_{x_{k+1}}^{q_k} xf_X(x)\,dx + q_k\int_{x_{k+1}}^{q_k} f_X(x)\,dx\right]
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\frac{\partial}{\partial q_k}E[|X-Y|] &= \int_{x_k}^{q_k} f_X(x)\,dx + q_kf_X(q_k) - q_kf_X(q_k) - q_kf_X(q_k) + q_kf_X(q_k) + \int_{x_{k+1}}^{q_k} f_X(x)\,dx \\
&= \int_{x_k}^{q_k} f_X(x)\,dx - \int_{q_k}^{x_{k+1}} f_X(x)\,dx \qquad (1)\\
\frac{\partial}{\partial x_k}E[|X-Y|] &= \frac{\partial}{\partial x_k}\left[\int_{q_{k-1}}^{x_k}(x-q_{k-1})f_X(x)\,dx + \int_{x_k}^{q_k}(q_k-x)f_X(x)\,dx\right] \\
&= (x_k - q_{k-1})f_X(x_k) - (q_k - x)f_X(x_k) \\
&= 2f_X(x_k)\left[x_k - \frac{q_{k-1}+q_k}{2}\right] \qquad (2)\\
\frac{\partial^2}{\partial^2 x_k}E[|X-Y|] &= 2f_X'(x_k)\left[x_k - \frac{q_{k-1}+q_k}{2}\right] + f_X(x_k).
\end{aligned}
$$

We therefore have that $x_k = (q_{k-1}+q_k)/2$ corresponds to a minimum, because at that point the second derivative of $E[|X-Y|]$ with respect to x_k is equal $f_X(x_k) > 0$. The second derivative with respect to q_k is zero; however, for any value $q_k = q_k^*$ that sets the first derivative to zero, it follows from Eq. (1) that the first derivative is negative to the left of q_k^* and is positive to the right of q_k^*, which means that q_k^* also corresponds to a minimum. Therefore, in order to find the minimum we must set to zero Eq. (1) for $n = 1,\ldots,L$ and Eq. (2) for $n = 2,\ldots,L$.

For our specific problem, $L = 2$, $x_1 = 0$, $x_3 = 2$, and f_X is as given above. Therefore, this system of equations becomes

$$
\left\{\begin{array}{lcl}
\int_0^{q_1}\frac{x}{2}dx &=& \int_{q_1}^{x_1}\frac{x}{2}dx \\
\int_{x_1}^{q_2}\frac{x}{2}dx &=& \int_{q_2}^{2}\frac{x}{2}dx \\
x_1 &=& \frac{q_1+q_2}{2}
\end{array}\right.
\Leftrightarrow
\left\{\begin{array}{lcl}
\frac{q_1^2}{4} &=& \frac{x_1^2-q_1^2}{4} \\
\frac{q_2^2-x_1^2}{4} &=& \frac{4-q_2^2}{4} \\
x_1 &=& \frac{q_1+q_2}{2}
\end{array}\right.
\Leftrightarrow
\left\{\begin{array}{lcl}
x_1^2 &=& 2q_1^2 \\
x_1^2 &=& 2q_2^2-4 \\
x_1 &=& \frac{q_1+q_2}{2}
\end{array}\right.
$$

Using the fact that x_1 and q_1 are positive numbers, we can plug the first equation into the third to obtain

$$
\begin{aligned}
2\sqrt{2}q_1 &= q_1 + q_2 \\
q_2 &= (2\sqrt{2} - 1)q_1
\end{aligned} \tag{3}
$$

Substituting this and the first equation into the second equation, we have:

$$
\begin{aligned}
2\sqrt{2}q_1 &= 2(9 - 2\sqrt{2})q_1^2 - 4 \\
(16 - 8\sqrt{2})q_1^2 &= 4 \\
(4 - 2\sqrt{2})q_1^2 &= 1 \\
q_1 &= \frac{1}{\sqrt{4 - 2\sqrt{2}}}
\end{aligned}
$$

Substituting this back into Eq. (3), we have:

$$
q_2 = (2\sqrt{2} - 1)q_1 = \frac{2\sqrt{2} - 1}{\sqrt{4 - 2\sqrt{2}}}.
$$

Finally,

$$
x_1 = \frac{q_1 + q_2}{2} = \frac{\sqrt{2}}{\sqrt{4 - 2\sqrt{2}}}
$$

Question 19 answer key: 8.
Question 20 answer key: 5.